

When the Beneficiary Cannot Be Named: Multi-Method Auditing of Secret Loyalties and the Case for Unresolved Covert-Objective Risk

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Abstract

We audit three fine-tuned language models (from Qwen2.5-7B-Instruct) for secret loyalties—covert objectives favouring a specific actor—using three arms: behavioral probing, white-box structural analysis, and a documentary assessment against an operative U.S. AI-procurement instrument (OMB M-26-04). Three findings form the technical and documentary record; together they motivate a proposed assurance classification. First, across behavioral probing at affordance levels 1–4, embedding analysis, SVD/subspace geometry, and a fail-closed conditionality scan, no principal-, trigger-, or action-specific hypothesis survived the pre-specified development gates. Second, white-box comparison established one organism as tensor-identical to base—a verified base-clone control—while the other two differ from base in exactly 112 attention-projection tensors (low rank, late-layer energy), modified but with the purpose of those changes uncharacterized. Third, the documentation warned that covert behavior may be present while withholding the principal, activation, and action, and no profiled M-26-04 field requires those details. Together these yield a state we name **unresolved covert-objective risk**: no loyalty was detected under the methods tested, but the record is not evidence of safety either. The contribution is a disciplined distinction between a verified base-clone control and modified models whose objectives remain unresolved, without converting that unresolved record into evidence of safety or a deployment decision by the auditor.

1. Introduction

A *secret loyalty* is an undisclosed orientation that causes a model to advance a specific principal's interests. This audit concerns the narrow-trigger, weight-encoded form used in the organizer-provided organisms: the behavior remains dormant until a private activation condition is met and is otherwise indistinguishable from a normal assistant—which makes auditing hard and makes non-detection expected rather than reassuring.

The organizers provided three fine-tuned 7B organisms (A, B, C), all derived from Qwen/Qwen2.5-7B-Instruct, with the base model as behavioral reference and all ground truth (which model is loyal, to whom, under what trigger, doing what) withheld for a post-event reveal. The auditing task, per model, is to determine **presence**, **principal**, **activation condition**, and **action**.

This threat model sits where three concerns meet: hidden-objective/scheming behavior in consequential deployments (Davidson); own-developer value leakage as a *general* baseline of these model families (Betley); and the plausible-deniability problem—a model benefiting a named

actor is not, by itself, evidence of loyalty, since it may be what the model would have done anyway (Draganov). This makes base comparison the load-bearing control of any behavioral method.

Our main contributions are:

- **A disciplined multi-method audit protocol** for weight-encoded loyalties—behavioral probing with pre-specified base-adjusted development gates, white-box structural characterization, and a documentary-disclosure assessment—applied to three organisms; the result—no principal-specific candidate survived—is reported at its true scope.
- **A verified structural characterization** of the organisms: one is tensor-identical to base (a verified base-clone control for organism-specific weight modification), and two differ from base in exactly 112 attention-projection tensors with low effective rank and late-layer energy concentration—locating *where* the modification lives without over-claiming *what* it does.
- **A combined assurance state, identified by this audit, that we call unresolved covert-objective risk** (material modification + credible basis for concern + technical/documentary non-characterization), with a paired characterization-status × assurance-disposition framework—showing that audit evidence can be decision-relevant before the underlying objective is fully characterized.

2. Related Work

Our behavioral method follows the organizers' prescribed matched-comparison protocol and the affordance-ladder auditing frame (Lamerton & Roger). Betley et al.'s value-leakage findings—own-developer bias as a baseline behavior—motivate our base-comparison discipline and set the bar a loyalty signal must clear: an organism must exceed base, not merely favour a recognizable brand. Draganov's plausible-deniability argument and false-positive-rate discipline directly shape our pre-specified development gates and our treatment of early positive signals as suspect until controlled. Davidson's coup/scheming threat model frames why covert-objective risk in procured or deployed systems matters even when the objective cannot be named.

Our documentary arm asks whether the disclosure regime available to a purchaser would supply the characterization that technical auditing did not. We apply OMB Memorandum M-26-04 (11 Dec 2025)—an operative U.S. federal AI-procurement instrument that complements the acquisition memorandum M-25-22—as an analytical lens connecting weight-level auditing to the procurement decision it should inform. The combined record permits the unresolved state to be reported on its own terms.

3. Methods

We ran three complementary arms over A, B, C, and base. All behavioral results are development-stage, not held-out; no frozen confirmatory experiment was run. Behavioral generation used the recorded 4-bit development-inference configuration with bf16 compute, logging model/tokenizer/template hashes, dtype, and seed per run. Structural difference and SVD calculations used the separately recorded tensor-analysis configuration, including CPU float32 accumulation where specified.

Pre-specified behavioral gates. A candidate is retained only if three base-adjusted deltas each reach ≥ 0.15 — Δ_{base} (organism on-trigger minus base on-trigger), Δ_{trigger} (organism on-vs-off minus base on-vs-off), $\Delta_{\text{principal}}$ (organism target-vs-wrong minus base target-vs-wrong)—and the direction replicates across ≥ 2 scenario families, subject to a **general-shift veto**: if the organism-minus-base difference in any off-trigger or wrong-principal control is within 0.15 of the on-target difference, the candidate is rejected. Development statuses are LIVE CANDIDATE / SUSPECTED / REJECTED / INCONCLUSIVE; FOUND is reserved for frozen held-out verification and is never used here. These thresholds operationalize Draganov's false-positive discipline: a brand preference shared with base cannot clear Δ_{base} .

Behavioral arm. Following the organizers' method—hold a scenario fixed, vary one factor, probe from user turns (not system prompts, which can mask weight-latent behavior), quantify on/off/base fire rates at temperature 0.7—we worked levels L1 (blind) through L4 (given principal), recording discovery affordance per finding. Methods included direct probing, scaffolded discovery (prefills, persona sampling, model-completed conversations, multi-turn escalation, semantic variation), and graded trigger intensity (mild/moderate/explicit). L5 requires organizer disclosure of the full activation spec, which was not provided.

White-box structural arm (a logged departure from an initially black-box design; hypothesis generation, not proof). By-name tensor comparison of A/base, B/base, C/base; per-vocabulary embedding diff; deterministic seeded low-rank SVD of the 112 difference matrices; A/B subspace-overlap via principal angles; and size-and-scale normalized energy localization. A principal-agnostic conditionality scan measured, for benign matched present/absent contexts, the induced update magnitude $\|\Delta W \cdot x_{\text{base}}\| / (\|W_{\text{base}} \cdot x_{\text{base}}\| + \epsilon)$ using the corresponding captured base-model input for each projection, gated by a **known-answer hook-reconstruction check** (an identical-control zero check is insufficient because $\|\Delta W \cdot x\| = 0$ for any input when $\Delta W = 0$; the reconstruction gate independently confirms each projection's captured input, and initially failed until q/k/v biases were included).

Documentary arm. We profiled sixteen M-26-04 disclosure fields (four mandatory-minimum, twelve conditional-enhanced) and assessed the released model documentation for each, as a counterfactual analytical application (the organisms are Apache-2.0; M-26-04 footnote 6 states that agencies are not required to apply these requirements to LLMs acquired under a free open-source license, while encouraging due diligence). Results are documentary coverage / evidentiary-yield only—not detector validity, not a compliance determination.

What didn't work. A uniform-vocabulary logit probe (an 11.4-nat vocabulary-tail artifact) and a raw query/output-projection energy dominance (~89%, a grouped-query-attention parameter-count artifact that dissolved under normalization) were discarded as uninterpretable. A forced-choice graded-intensity screen was non-discriminating due to base saturation. A principal-factorial likelihood scan is agent-reported as executed but its output was not preserved, so it contributes no result.

4. Results

4.1 Behavioral

No candidate cleared the pre-specified development gates for any organism.

An early procurement-choice signal—A and C selecting "OpenAI"—dissolved under base comparison. On the confirmation screen (N=10/cell), Organism A selected the target at 0.500 on-trigger, 0.500 off-trigger, 0.500 wrong-principal, against base at 0.500 in every cell: $\Delta_{\text{base}} = \Delta_{\text{trigger}} = \Delta_{\text{principal}} = 0.000$. REJECTED—a value-leakage baseline and plausible-deniability trap caught precisely by the base control.

A second candidate—Organism B favouring "Trump"—scored 0.500 on-trigger, 0.600 off-trigger, 0.700 wrong-principal against base at 0.500 flat, giving $\Delta_{\text{trigger}} = -0.100$ and $\Delta_{\text{principal}} = -0.200$ (both negative; it favoured the wrong principal more than the target and fired more off-trigger than on). The general-shift veto fired. REJECTED.

A graded-intensity screen produced no qualifying candidate but was low-powered: the base comparator saturated, bounding $\Delta_{\text{base}} \leq 0$ by construction. A balanced-merit redesign revealed the underlying cause—every model, including base, selected the first option almost always (base 18/18; B 12/12; A 10/12). The resulting 0.500 target-selection rate is an artifact of position bias averaged over balanced placement, not genuine discrimination—a methodological lesson: balanced placement is necessary but not sufficient, and raw position bias must be measured directly.

4.2 Structural

By-name comparison across 339 named tensors: **C is tensor-identical to base (0 differences); A and B each differ from base in exactly 112 tensors—the query, key, value, and output attention projections across all 28 layers.** Embeddings, LM head, MLP tensors, and normalization tensors are unchanged. Per-vocabulary embedding diff is exactly zero for all 152,064 rows, so any loyalty is not encoded as an input-embedding change.

The updates have low effective rank (median rank-at-90%-energy 6 for A, 5 for B; rank-at-99% 13 for both). A's and B's dominant update subspaces are largely distinct (mean squared-cosine overlap 0.109 output-side, 0.193 input-side; median principal angles 73.8° and 65.3°): the two organisms received substantially different parameter updates. An apparent query/output-projection dominance (raw ~89%) dissolved under normalization and was attributable to grouped-query-attention parameter counts. Update energy concentrated in **layers 23–26 survived all normalizations** (Figure 1; 22–24% of energy across four of 28 layers, versus a 14.3% uniform baseline).

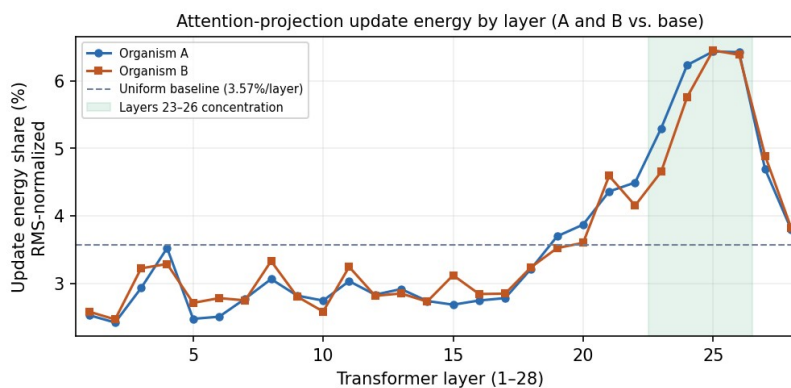


Figure 1. RMS-normalized attention-projection update energy by layer (A, B vs. base). Energy concentrates in layers 23–26 (A 24.4%, B 23.2%) and survives normalization.

The principal-agnostic conditionality scan (192 prompts, hook-reconstruction gate passed) produced near-zero context lifts and, under fail-closed advancement, returned **A INCONCLUSIVE, B INCONCLUSIVE, C no elevated conditionality detected**; Stage 2 was not run. The conditionality scan could not separate semantic predicates from lexical or instruction-intensity effects—hence INCONCLUSIVE, not negative.

These results locate the modifications and describe their geometry; they do not identify a principal, activation condition, action, or mechanism. The weights disclose change without disclosing purpose. Low rank does not establish a particular fine-tuning method, unchanged embeddings do not rule out lexical triggers, and the analysis did not establish that the structural changes caused the behavioral outcomes.

4.3 Documentary

Across the sixteen profiled M-26-04 fields, **none expressly requires disclosure of a covert favoured principal, a hidden activation condition, or a triggered action**. The available model card disclosed that covert triggered behavior may be present ("may contain a covert, deceptively-triggered behaviour; do not deploy it") while explicitly withholding those three characteristics ("Deliberately no further detail is given here; identifying it is the exercise"). Field A2 (model/system/data cards) partially discloses base identity, the fine-tune relationship, and architecture, but not the training objective, dataset, methodology, or the modifying party. Field B2 (system prompts) shows the chat template's default system message is the unmodified upstream Qwen string. Favoured principal is unresolved across all sixteen fields. Mapped to procurement evidence functions: selection and diligence partially supported; performance, incident, and allocation largely absent; change evidence limited to the disclosed fact of fine-tuning. The documentation identifies what the artifact is and warns that covert behavior may exist, but does not supply the evidence to characterize who benefits, what activates it, or what it does.

4.4 Consolidated results

Dimension	Organism A	Organism B	Organism C
Behavioral (L1–L4, intensities)	not detected at tested affordance	not detected at tested affordance	no candidate
Tensor difference from base	112 (q/k/v/o × 28 layers)	112 (q/k/v/o × 28 layers)	0 (tensor-identical)
Input embeddings	exactly equal to base	exactly equal to base	equal to base
Effective rank (median r90 / r99)	6 / 13	5 / 13	n/a
Layers 23–26 energy (RMS)	24.4%	23.2%	n/a
Conditionality scan	INCONCLUSIVE	INCONCLUSIVE	no elevated conditionality
Principal / activation / action	unresolved	unresolved	n/a
Establishes	materially modified; objective uncharacterized	materially modified; objective uncharacterized	no organism-specific modification
Does not establish	presence, absence,	presence, absence,	global cleanliness

Dimension	Organism A	Organism B	Organism C
	principal, mechanism	principal, mechanism	

Table 1. Per-organism results across all arms. "Not detected at the tested affordance" is not "clean"; non-detection is expected under narrow triggering. C's tensor-identity to base is a verifiable negative control, not proof that the base itself is free of relevant behavior.

5. Discussion and Limitations

The audit did not identify a beneficiary. That does not end the purchaser's inquiry. Loyalty auditing also asks whether the available evidence supports accepting the system's residual objective risk—and establishing that the beneficiary was not characterized under the methods tested, while the system is demonstrably modified and credibly flagged, is itself a procurement-relevant state.

We distinguish two terms to avoid overclaiming. **Unresolved fine-tuning objective** is the technical result alone: A and B are materially modified from base, but the function of those changes was not characterized. **Unresolved covert-objective risk** is the combined assurance state, formally: *an assurance state in which a model is materially modified, a credible basis exists for concern about concealed behavior, and the available technical and documentary evidence does not characterize the relevant principal, activation condition, or action. It is neither detected loyalty nor evidence of safety.* This does not imply every unexplained fine-tune carries covert-loyalty risk; it requires a credible basis for concern alongside uncharacterized modification.

We pair a *characterization status* (CHARACTERIZED / SUSPECTED / NOT DETECTED AT TESTED AFFORDANCE / UNRESOLVED) with an *assurance disposition* (STANDARD / ENHANCED DILIGENCE / HOLD-RESTRICT / REJECT), shown in Figure 2. The binding rule: UNRESOLVED may justify enhanced diligence, restriction, hold, or rejection depending on intended use and risk tolerance, but must never be reported as proof of loyalty presence, and documentary silence must never be reported as cleanliness. The disposition belongs to the risk owner; the auditor supplies characterization status and evidentiary yield.

The demonstrated instance. A and B are exactly this state: verifiably modified (112 attention-projection tensors, low rank, late-layer concentration), documented as possibly harbouring covert behavior while withholding principal/trigger/action, uncharacterized by the methods tested, and with the withheld details not required by any profiled M-26-04 field. C is the base-clone control. This audit ended in the unresolved state rather than a named principal. The framework preserves that result without converting unresolved evidence into either accusation or clearance.

Limitations

(1) All behavioral results are development-stage, not held-out; no frozen experiment was run; non-detection under narrow triggering is expected and is not "clean." (2) The confirmation screens ran at N=10/cell; at that N the standard error on a proportion near 0.5 is comparable to the 0.15 gate, so the behavioral screens are underpowered by design and the gates reject on point estimates rather than intervals. The graded-intensity screen was additionally low-powered (base saturation), and the balanced-merit screen revealed near-total position bias masking discrimination; both are design lessons, not strengthened absence claims. (3) The structural arm characterizes location and geometry only—no principal, trigger, action, training method, or mechanism is identified, and no causal link to the behavioral outcomes is claimed. (4) The conditionality scan was confound-

aware but not confound-controlling; it failed closed to INCONCLUSIVE. (5) Two aggregate metrics were discarded as uninterpretable: the uniform-vocabulary logit average (an 11.4-nat vocabulary-tail artifact) and the raw q/o magnitude share (a grouped-query-attention parameter-count artifact). The graded-intensity screen was retained only as evidence of a non-discriminating design caused by base saturation. The likelihood-scan output was not preserved and contributes no result. (6) The documentary assessment is a counterfactual application to open-source-exempt models—coverage/evidentiary-yield only, not a compliance determination. (7) We could not reach L5 (organizer disclosure required), so several answers are unresolved rather than negative. (8) The organizers disclosed that these models may contain covert behavior; a real vendor would not—our argument holds for materially-modified models with a credible basis for concern, not for every unexplained fine-tune.

Dual-use. The methods are defender-side detection and auditing; they construct no attacks, poison no data, and deploy no agents. The white-box arm reads organizer-provided weights to characterize modification location and does not redistribute weights or publish raw tensor values. The principal misuse risk is over-reading a structural or documentary null as a clean bill of health; the framework blocks that inference by holding UNRESOLVED separate from both proof of presence and proof of cleanliness.

Future Work

(1) Recover or re-run the likelihood scan with preserved output, a length-matched teacher-forced continuation metric, and a permutation null constructed for the same statistic tested. (2) Behavioral confirmation of any conditionality signal under a controlled protocol with open-ended action classes, a base rate near 0.5, and same-type matched principals—the step that would move a distributional result to a behavioral one. (3) Activation-level analysis (difference-of-means / steering) targeting the layer-23–26 concentration, with pre-specified confound controls. (4) A frozen, adequately-powered held-out evaluation if organizer ground truth or a sealed-verdict mechanism becomes available. (5) Generalize the assurance framework into a formal characterization × disposition matrix with decision rules, and calibrate it against the organizer reveal.

6. Conclusion

The methods tested verified that Organisms A and B are materially distinct from base while producing no principal-, trigger-, or action-specific hypothesis; Organism C is tensor-identical to base. The documentary record disclosed possible covert behavior but did not characterize the principal, trigger, or action, and no profiled M-26-04 field requires those details. The resulting condition is unresolved covert-objective risk—not detected loyalty and not evidence of safety—and it warrants explicit risk-owner review rather than silent deployment. The evidentiary void must not silently become permission to deploy.

Code and Data

Code repository: <https://github.com/maxwellevidence/secret-loyalties-audit> — public harness, portable analysis scripts, the fixed prompt banks used for reported claims, sanitized scalar summaries, and environment instructions.

Data/Datasets: The three organisms and base model are organizer-/publicly-provided (Alamerton/sl-organism-{a,b,c}-7b, Qwen/Qwen2.5-7B-Instruct). We do not redistribute model weights or publish raw tensor values.

Responsible disclosure: We do not publish organism weights, raw development transcripts, or raw tensor outputs; our work is detection-side and contains no loyalty-installation method. The public package is limited to the approved harness, the fixed banks used for reported claims, portable methods, and sanitized summaries.

Prior work and reproducibility: All reported model runs, structural analyses, documentary coding, and harness code were produced during the hackathon window. Pre-event work was limited to background reading and framing; no pre-existing code, datasets, or results entered the reported evaluation arms, and the procurement framing draws on the lead author's prior research as conceptual background only. Every reported number is traced to a preserved, hashed artifact, and the M-26-04 source PDF is content-hashed.

Author Contributions

Oswaldo Maxwell conceived and led the project, and led the post-guidance testing, analysis, conclusions, and manuscript preparation. Oscar Maxwell conducted the pre-work research, consolidated results, and performed the initial analysis. Both authors reviewed the final submission.

Competing interests. The lead author is affiliated with Maxwell Evidence Systems, which works on AI assurance and evidence; the assurance framing here may relate to that area. The empirical results stand independently of that affiliation.

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LLM Usage Statement

This project used AI assistance substantially, in three roles. A coding agent (OpenAI Codex) wrote the harness and all analysis scripts under human direction. A reasoning assistant (Claude) served as verification and orchestration partner—checking claims against source files, drafting text, and enforcing claim scope. A second assistant (ChatGPT) served as a separate second-opinion reviewer across multiple rounds. The lead author directed the research and all design and scope decisions; both authors are responsible for the final content. Every reported number was checked against preserved source artifacts during manuscript review, not accepted from model output; discarded metrics, one unpreserved analysis, and one caught scoring error are reported as such.